Schrödinger in Spacekime: Formulating and Solving the Schrödinger Equation with Complex Time

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Abstract

We extend the 4D spacetime model to 5D spacekime, in which time is a complex number $\boldsymbol{t = a + bi}$ with both a real component that orders events in spacetime and an imaginary component that constrains possible events. Complex time (kime) can be visualized as a two-dimensional area in which "thickness" along real time corresponds to possibility (incomplete specification), while complete specification locally narrows complex time to a line. Here we investigate the implications of extending the Schrödinger equation to complex time, allowing separate evolution rules in the real time and imaginary time directions. We maintain unitary dynamics in real time but dissipative dynamics in imaginary time, which leads to an effective non-Hermitian Hamiltonian $\boldsymbol{H_1 + iH_2}$ over a selected contour in the complex plane. This yields a more general wavefunction $\boldsymbol{\psi(x, a, b)}$ containing probabilities of all possible remaining spacetime states. The norm of the wavefunction is no longer constant during Schrödinger evolution, describing an open quantum system with continuous collapse. However, the normalized observable wavefunction can always be recovered by marginalizing over imaginary time support. We can further decompose $\boldsymbol{H_2}$ as a sum of two operators $\boldsymbol{H_2 = H_L + H_E}$, where $\boldsymbol{H_L}$ produces maximum-entropy realizations of the imaginary-time support (law-like dynamics) and $\boldsymbol{H_E}$ provides a "reality dial" for external inputs to the system. Complex-time Schrödinger thus provides an elegant description of continuous wavefunction collapse, which occurs when $\boldsymbol{H_E}$ causes imaginary time thickness to approach 0. The Born rule represents proper bookkeeping over imaginary time support. The eigenvalues of the Hamiltonian are complex numbers in which the real part orders states by energy in spacetime and the imaginary part orders states by ontological persistence in imaginary time.

The conventional Schrödinger equation can now be seen as the minimal dynamics that preserve coherence over imaginary time support. Complex-time Schrödinger evolution allows non-Markovian dynamics and feedback, and thus is strictly more general than the Lindblad equations, but reduces to Lindblad evolution under additional restrictions. Thus, the complex-time Schrödinger equation provides new descriptions of complex environmental interactions in open quantum systems such as quantum computers.
Complex-time Schrödinger dynamics also suggest a new semiclassical theory of quantum gravity in which Einstein's spacetime stress-energy tensor arises from integrating over the imaginary-time support of a wavefunction. Intuitively, our insight is that proper stress-energy bookkeeping in spacetime requires considering partially collapsed wavefunctions, which can still gravitate. Our theory reduces to general relativity in the limit of complete wavefunction collapse, and the Bianchi rule still holds exactly in spacetime. Our approach accounts for the results of the classic Page-Geilker gravitational experiment; allows gravitational entanglement; and yields predictions that can be tested in the near term with recently proposed gravitational superposition and gravitational entanglement experiments.
Imaginary-time thinning during wavefunction collapse can also be viewed as a fluid flow into spacetime that may explain dark matter and dark energy. Positing a conservation law for this "fluid" implies a kinematic relation between cosmic expansion and imaginary-time thinning. Thus, the remaining depletable imaginary-time measure $\mathbf{\Theta}$ acts as a single structural variable whose regimes correspond to dark matter (partially collapsed wavefunction), baryonic matter (fully collapsed wavefunction), and dark energy (irreducible minimum quantum of imaginary-time support). We derive a minimal model driven by complex-time Schrödinger dynamics and show that the observed present-day dark-to-baryonic matter ratio constrains the effective thinning rate to be a fraction of the Hubble rate over cosmic timescales. The same thinning dynamics determine both gravitational attraction (through spatial variation of $\mathbf{\Theta}$) and expansion (through global thinning), yielding a dimensionless balance parameter controlling structure formation. We further demonstrate that the framework is compatible with collisionless dark matter behavior in galaxy mergers. Our model allows, but does not require, dark matter particles such as WIMPs or axions. The proposal thus offers a unified ontological substrate for the dark sector in terms of complex-time Schrödinger evolution. Crucially, our proposal remains fully consistent with established cosmological phenomenology and providing operator-level constraints on complex-time Schrödinger evolution.

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1 Formulating and Solving the Schrödinger Equation in Spacekime

1.1 Introduction to Spacekime

We begin by introducing the notion of spacekime, in which conventional spacetime is augmented by treating time as a complex number. That is: $t = a + bi$, where a and

b are real numbers and $i = \sqrt{-1}$. We refer to this augmented time as complex time or kime. The real component of kime a represents the familiar temporal ordering of events in spacetime–what actually happens, causally and sequentially. The imaginary component of kime b, by contrast, does not correspond to an additional temporal dimension along which events unfold. Instead, it encodes possibility structure: constraints on which possible sequences of events across time can be jointly realized in a single, globally consistent world history. Real time orders events, supports causation, and is asymmetric. Imaginary time encodes symmetry, encodes periodicity, and encodes possibility structure rather than occurrence.

Thus, in spacekime, real time describes what *actually* happens, and imaginary time describes all possible imaginary events that *could* happen. This distinction motivates the notion of spacekime thickness, which reflects how extensively a system's behavior is constrained across imaginary time: thick spacekime structure corresponds to robust coherence among counterfactual alternatives, whereas thin spacekime structure corresponds to effective restriction to a narrow family of histories, yielding behavior that appears classical and localized.

To explore the concept of spacekime mathematically,we solve the Schrödinger equation with a time-independent potential $V(\vec{r})$ using a Cartesian parameterization of complex time $t = a + ib$, where a is the real part (corresponding to the oscillatory component) and b is the imaginary part (introducing diffusive effects). This is equivalent to the polar form $t = \tau e^{i\phi}$ via $a = \tau \cos\phi$ and $b = \tau \sin\phi$, with $\tau = \sqrt{a^2 + b^2}$ and $\phi = \tan^{-1}(b/a)$, but we'll work directly in (a, b). For simplicity, use units so that $\hbar = 1$, $m = 1$. We use a 3D isotropic harmonic oscillator potential $V(\vec{r}) = \frac{1}{2}\omega^2|\vec{r}|^2$.

1.2 Conventional Schrödinger Equation

The time-dependent Schrödinger equation with potential is:

$$i\frac{\partial\psi(\vec{r},t)}{\partial t} = \left(-\frac{1}{2}\nabla^2 + V(\vec{r})\right)\psi(\vec{r},t) = \hat{H}\psi(\vec{r},t) \tag{1}$$

with initial condition $\psi(\vec{r},0) = \psi_0(\vec{r})$.

The formal solution is $\psi(\vec{r},t) = e^{-i\hat{H}t}\psi_0(\vec{r})$, expressed via the propagator $K(\vec{r},\vec{r}',t)$:

$$\psi(\vec{r},t) = \int K(\vec{r},\vec{r}',t)\psi_0(\vec{r}')\, d^3r' \tag{2}$$

1.3 Complex-Time Schrödinger Dynamics Without Holomorphy

We consider a generalized time coordinate $t = a + ib$, where a represents ordinary real time and b represents an imaginary-time fiber coordinate associated with remaining unrealized possibilities. We do not assume that the wavefunction is holomorphic in t. Instead, the fundamental object of the theory is a field

$$\psi(x, a, b) \tag{3}$$

defined over spacetime together with an imaginary-time fiber. Thus we allow dynamics to specified independently along the two directions in complex time $t = a + ib$.

Evolution along the real-time direction is unitary and generated by a Hermitian Hamiltonian H_1:

$$i\hbar\, \partial_a \psi(x, a, b) = H_1\, \psi(x, a, b). \tag{4}$$

Evolution along the imaginary-time fiber represents the thinning of unrealized possibilities and is generated by a Hermitian operator H_2,

$$\hbar\, \partial_b \psi(x, a, b) = -H_2\, \psi(x, a, b), \tag{5}$$

where H_2 may contain contributions from outside the system, such as constraints, environmental interactions, or measurement/observation.

Physical evolution thus corresponds to following a contour in the complex plane along which real time advances as imaginary-time support thins. More formally, we can specify a gauge condition so that the contour enforces monotonic imaginary-time thinning in real time:

$$\frac{da}{db} = -\frac{1}{v(a, b)}, \qquad v(a, b) > 0. \tag{6}$$

Along this contour the physical state is

$$\psi_{\text{phys}}(x, b) = \psi(x, a(b), b). \tag{7}$$

Applying the chain rule,

$$\frac{d}{db}\psi_{\text{phys}} = \partial_b \psi + \frac{da}{db}\partial_a \psi. \tag{8}$$

Substituting the two dynamical generators yields

$$i\hbar\, \frac{d}{db}\psi_{\text{phys}} = \left(\frac{1}{v(a, b)} H_1 + iH_2\right) \psi_{\text{phys}}. \tag{9}$$

Thus the combined evolution along the contour can be expressed in terms of an effective non-Hermitian generator,

$$H_{\text{eff}} = \frac{1}{v(a, b)} H_1 + iH_2. \tag{10}$$

The extended state $\psi(x, a, b)$ represents an amplitude density over possible realizations indexed by imaginary-time support b. The observable quantum state is obtained by marginalizing over the fiber,

$$\Psi(x, a) = \int_{\mathcal{F}_b} W(b)\, \psi(x, a, b)\, db, \tag{11}$$

where $W(b)$ is a linear weighting kernel determined by a stochastic quantization process (see below for details).

The ordinary Schrödinger equation for $\Psi(x, a)$ emerges directly by differentiating the marginal state and substituting the a-dynamics. To make this explicit, allow the admissible fiber support to thin with real time,

$$\Psi(x, a) = \int_{b_-(a)}^{b_+(a)} W(b)\, \psi(x, a, b)\, db. \tag{12}$$

Applying Leibniz' rule gives

$$\partial_a \Psi = \int_{b_-}^{b_+} W(b)\, \partial_a \psi\, db + \dot{b}_+\, W(b_+)\, \psi(x, a, b_+) - \dot{b}_-\, W(b_-)\, \psi(x, a, b_-). \tag{13}$$

Using $i\hbar\, \partial_a \psi = H_1 \psi$ and assuming H_1 acts on x (not on b),

$$i\hbar\, \partial_a \Psi = H_1 \int_{b_-}^{b_+} W(b)\, \psi\, db + i\hbar\, \mathcal{B}(x, a) = H_1 \Psi + i\hbar\, \mathcal{B}(x, a), \tag{14}$$

where

$$\mathcal{B}(x, a) = \dot{b}_+\, W(b_+)\, \psi(x, a, b_+) - \dot{b}_-\, W(b_-)\, \psi(x, a, b_-) \tag{15}$$

is the net boundary flux from the imaginary-time fiber.

Therefore, if the support is fixed or boundary amplitudes vanish so that $\mathcal{B} = 0$, we recover the standard real-time Schrödinger equation

$$i\hbar\, \partial_a \Psi = H_1 \Psi. \tag{16}$$

When $\mathcal{B} \neq 0$ (thinning, selection, measurement), the projected dynamics become

$$i\hbar\, \partial_a \Psi = (H_1 - i\Gamma)\, \Psi, \tag{17}$$

with Γ the effective non-Hermitian term induced by fiber-boundary flux. Because both the marginalization operator and the b-dynamics are linear in ψ, linear superposition of states is preserved.

In summary, $\psi(x, a, b)$ is an amplitude density over possible realizations; b represents remaining imaginary-time support; $\Psi(x, a)$ is the realized physical wavefunction; collapse corresponds to shrinking support in b; decoherence corresponds to diffusion in b; and measurement corresponds to boundary conditions imposed on the fiber.

1.4 Integrability of Complex-Time Dynamics

Because the state $\psi(x, a, b)$ evolves independently along the real-time direction a and the imaginary-time fiber direction b, consistency of the theory requires that the resulting two-dimensional evolution be well defined. In particular, the state obtained by evolving first in a and then in b should agree with the state obtained by evolving in the opposite order, unless explicit path dependence is intended.

The real-time and imaginary-time dynamics are

$$i\hbar\,\partial_a\psi = H_1\,\psi, \tag{18}$$

$$\hbar\,\partial_b\psi = -H_2\,\psi, \tag{19}$$

where H_1 and H_2 are Hermitian operators that may depend on (x, a, b).

Requiring equality of mixed partial derivatives,

$$\partial_a\partial_b\psi = \partial_b\partial_a\psi, \tag{20}$$

imposes a compatibility constraint on the generators. Substituting the dynamical equations gives

$$\partial_a(-H_2\psi) = \partial_b\left(\frac{-i}{\hbar}H_1\psi\right). \tag{21}$$

After rearranging terms one obtains the condition

$$\partial_a H_2 - \partial_b H_1 + \frac{i}{\hbar}[H_1, H_2] = 0. \tag{22}$$

This equation is analogous to a zero-curvature or flat-connection condition in gauge theory. When it is satisfied, the evolution in the (a, b) plane is path independent, and the state $\psi(x, a, b)$ is uniquely defined throughout the extended manifold once initial data are specified.

If H_1 and H_2 do not depend explicitly on a or b, the integrability condition reduces to

$$[H_1, H_2] = 0. \tag{23}$$

In this case the real-time unitary dynamics and the imaginary-time thinning dynamics commute, and the order of evolution along a and b is irrelevant.

More generally, the generators need not commute. In this case the evolution in the (a, b) plane becomes path dependent. Physically, this means that the interaction between unitary dynamics and thinning dynamics can depend on the history of constraint enforcement in the imaginary-time fiber.

The physical evolution considered in this work corresponds to a preferred contour in the (a, b) plane along which imaginary-time support monotonically thins while real time advances. Along such a contour the dynamics are governed by the effective non-Hermitian generator derived in Sec. 1.3. Path dependence away from this contour can be interpreted as reflecting the nontrivial structure of the constraint dynamics acting on the imaginary-time fiber.

The structure of Eq. (13) closely resembles the compatibility conditions arising in stochastic quantization, where physical observables emerge as stationary solutions of a fictitious-time evolution [1]. In the present framework, however, the imaginary-time coordinate b represents a finite fiber of remaining possibilities rather than an auxiliary equilibration parameter.

1.5 Complex-Time Solution for 3D Isotropic Harmonic Oscillator

We next demonstrate how to solve the complex-time Schrödinger equation for a 3D isotropic harmonic oscillator. The potential and Hamiltonian are:

$$V(\vec{r}) = \frac{1}{2}m\omega^2|\vec{r}|^2, \qquad H_{\text{osc}} = -\frac{\hbar^2}{2m}\nabla^2 + \frac{1}{2}m\omega^2|\vec{r}|^2,$$

Because ψ is not holomorphic, we cannot solve by analytic continuation $t \mapsto a + ib$. Instead, we solve the two evolution equations independently:

$$i\hbar\,\partial_a\psi = H_1\psi, \tag{24}$$

$$\hbar\,\partial_b\psi = -H_2\psi. \tag{25}$$

As a solvable thinning model consistent with Sec. 1.3, choose

$$H_1 = H_{\text{osc}}, \qquad H_2 = -\lambda H_{\text{osc}}, \qquad \lambda > 0,$$

so $[H_1, H_2] = 0$ and the two flows commute.

Let $\phi_{\mathbf{n}}(\vec{r})$ be 3D oscillator eigenfunctions with $E_{\mathbf{n}} = \hbar\omega(N + \frac{3}{2})$, $N = n_x + n_y + n_z$, and expand $\psi_0(\vec{r}) = \sum_{\mathbf{n}} c_{\mathbf{n}}\phi_{\mathbf{n}}(\vec{r})$. Then

$$\psi(\vec{r}, a, b) = \sum_{\mathbf{n}} c_{\mathbf{n}} \exp\left[-\frac{i}{\hbar}E_{\mathbf{n}}(a - a_0)\right] \exp\left[\frac{\lambda}{\hbar}E_{\mathbf{n}}(b - b_0)\right] \phi_{\mathbf{n}}(\vec{r}). \tag{26}$$

The gauge condition

$$\frac{da}{db} = -\frac{1}{v(a, b)}, \qquad v(a, b) > 0,$$

implies $db/da = -v < 0$: imaginary-time support thins monotonically as real time advances. Define

$$s(a) := b_0 - b(a) = \int_{a_0}^{a} v(a', b(a'))\; da' \geq 0.$$

Along the physical contour,

$$\psi_{\text{phys}}(\vec{r}, a) = \sum_{\mathbf{n}} c_{\mathbf{n}} \exp\left[-\frac{i}{\hbar}E_{\mathbf{n}}(a - a_0)\right] \exp\left[-\frac{\lambda}{\hbar}E_{\mathbf{n}}s(a)\right] \phi_{\mathbf{n}}(\vec{r}). \tag{27}$$

Note that this representation has no branch-cut ambiguity because no holomorphic continuation is assumed.

For the Gaussian ground state

$$\psi_0(\vec{r}) = \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} \exp\left(-\frac{m\omega}{2\hbar}|\vec{r}|^2\right) = \phi_{000}(\vec{r}),$$

only one mode contributes $\left(E_{000} = \frac{3}{2}\hbar\omega\right)$, so

$$\psi_{\text{phys}}(\vec{r}, a) = \exp\left[-i\frac{3}{2}\omega(a - a_0)\right] \exp\left[-\frac{3}{2}\lambda\omega s(a)\right] \phi_{000}(\vec{r}). \tag{28}$$

For a general isotropic Gaussian width, the state expands over even oscillator modes, and each mode acquires the same phase factor in a and thinning factor in $s(a)$ shown above.

When $[H_1, H_2] = 0$, we obtain separable evolution in real and imaginary time because the two-parameter propagator factors as

$$U(a, b; a_0, b_0) = \exp\left[-\frac{i}{\hbar}H_1(a - a_0)\right] \exp\left[-\frac{1}{\hbar}H_2(b - b_0)\right]. \tag{29}$$

Under the thinning gauge $b = b(a)$ and the oscillator choice $H_2 = -\lambda H_1$,

$$U_{\text{phys}}(a) = \exp\left[-\frac{i}{\hbar}H_1(a - a_0)\right] \exp\left[-\frac{\lambda}{\hbar}H_1 s(a)\right]. \tag{30}$$

Thus real time a controls oscillatory phase evolution, while monotone decrease of b controls modal thinning through $s(a)$.

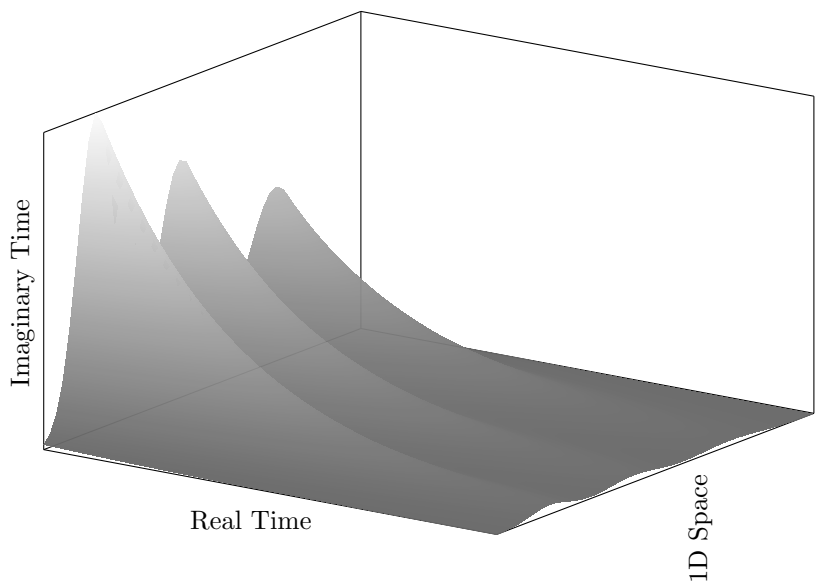

Fig. 1 Drawing of spacekime viewed from the outside. Each point in space has an associated imaginary time thickness indicating the amount of unconstrained possibility. As real time proceeds, imaginary time support remains constant or thins but never thickens. Spacetime "comes into focus" (indicated by increasing opacity) as thinning imaginary time support removes possibilities. Note that in this diagram, space is shown as a single dimension for easy depiction; spacekime is actually a 5-dimensional object.

Fig. 2 A schematic of spacekime thickness (imaginary-time support) tapering to a minimum along real time.

2 Spacetime as an Open Quantum System in Spacekime

We can now develop a new interpretation of quantum mechanics in which spacetime is an open system that reflects changeable possibilities in imaginary time.

We posit that physical spacetime (M) is augmented by a modal fiber at each spacetime point (x, t), where $a = \Re(t)$ is the real part of complex time $t = a + ib$. This fiber is an imaginary-time coordinate

$$b \in \mathcal{T}_{(x,t)} := 0, \Delta b, 2\Delta b, \ldots, \beta(x, t),$$

where: $\Delta b > 0$ is a fundamental quantum of imaginary time and $\beta(x, t)$ is a finite, possibly dynamical, upper bound. Therefore $\mathcal{T}_{(x,t)}$ is therefore a finite, discrete set.

Each $b_k \in \mathcal{T}_{(x,t)}$ corresponds to a distinct possible physical realization of the system at (x, t). Thus, imaginary time enumerates possibilities. This is an ontological claim, not merely a computational device.

All $b_k \in \mathcal{T}^*(x, t)$ are taken to be a priori equally admissible:

$$\mathbb{P}(b_k) = \frac{1}{|\mathcal{T}^*(x, t)|}.$$

However, possibilities indexed by different b_k are not independent. Without correlations across b, coherent superposition is generically destroyed by random phases. Therefore, additional structure is required to constrain how amplitudes associated with different b relate to one another.

This additional structure is supplied by a minimal set of constraints, formalized below.

2.1 Constraints on the imaginary-time fiber

Let $\Phi(x, a; b)$ denote the contribution to the physical state from the possibility indexed by b.

The set $\Phi(b)$ is not arbitrary. It must satisfy three fundamental constraints.

1. **Local coherence in imaginary time**. Contributions from neighboring values of b must be locally correlated:

$$\Phi(b + \Delta b) = \Phi(b) + O(\Delta b).$$

. Without this constraint, sums over b generically decohere.

2. **Maximum entropy**. Apart from physical constraints, there should be no additional structure in the imaginary-time direction. Formally, among all joint distributions over $\Phi(b)$ satisfying locality and physical constraints, the admissible ones are those of maximum entropy. Equivalently, the joint law of $\Phi(b)$ minimizes relative entropy with respect to a structureless reference measure. This essentially means minimal commitment to any open possibility at each point in spacetime.
3. **Constraint-driven suppression**. Physical context—Hamiltonian structure, environmental coupling, measurement setup, or external selection—enters through a constraint functional $S_E[\Phi]$. Configurations incompatible with S_E must be suppressed across the fiber, while compatible ones persist.

The unique minimal process satisfying local coherence, maximum entropy, and constraint-driven suppression is a Schrödinger-bridge or entropic-projection principle, called stochastic quantization in certain settings [1–3]. In continuum notation, the imaginary-time process takes the Langevin form

$$\partial_b \Phi = -\frac{\delta S_E[\Phi]}{\delta \Phi} + \eta(b), \qquad \langle \eta(b)\eta(b')\rangle = 2\nu, \delta(b-b'),$$

Equivalently, the probability law over fiber configurations minimizes

$$\mathrm{KL}(\mathbb{P}||\mathbb{P}_{\mathrm{Wiener}}) \quad \text{subject to} \quad \langle S_E \rangle = \text{const.}$$

Note that this stochastic quantization is not a physical time evolution, but rather a constraint on the allowed correlation structure across imaginary time.

2.2 The spacetime wavefunction as an integral over the fiber

We then define the physical wavefunction at spacetime point (x,t) as the coherent synthesis over the imaginary-time fiber:

$$\boxed{\psi(x,t) = \int_0^{\beta(x,t)} \Phi(x,a;b)db = \sum_{b\in\mathcal{T}_{(x,t)}} \Phi(x,a;b)}$$

in the continuum limit.

The resulting spacetime-facing state vector $|\Psi(a)\rangle$ need not be normalized. Define:

$$W(a) := \langle \Psi(a)|\Psi(a)\rangle, \qquad |\psi_{\mathrm{phys}}(a)\rangle := \frac{|\Psi(a)\rangle}{\sqrt{W(a)}} \quad (W>0).$$

Interpretation: $W(a)$ measures total admissible modal support. $|\psi_{\mathrm{phys}}(a)\rangle$ is the normalized physical state governing all conditional predictions. Probabilities should always computed from $|\psi_{\mathrm{phys}}\rangle$, not from $|\Psi\rangle$, taking into account the full range of open possibilities at a.

2.3 Emergence of the Spacetime Schrödinger Equation from Spacekime

We can now see how the spacetime Schrödinger equation arises from this spacekime model. Because ψ is defined as a coherent synthesis over the fiber, its physically meaningful content lies in transition probabilities

$$|\langle\phi|\psi\rangle|^2.$$

Between interventions that modify fiber constraints, real-time evolution (i.e., evolution along $a = \Re(t)$) must preserve this coherence structure. Therefore, free evolution must preserve inner products on rays.

By Wigner's theorem, any continuous, deterministic evolution on rays preserving transition probabilities is implemented by a unitary operator $U(a)$ [4]. By Stone's theorem [5],

$$U(a) = e^{-\frac{i}{\hbar}Ha}$$

for a unique self-adjoint operator H.

The physical state then satisfies

$$\boxed{i\hbar, \partial_a|\psi_{\text{phys}}(a)\rangle = H|\psi_{\text{phys}}(a)\rangle.}$$

Thus we can see that the Schrödinger equation is not an independent postulate. It is the unique minimal real-time dynamics compatible with a wavefunction defined as a coherent synthesis over an imaginary-time fiber with a Schrödinger bridge.

2.4 External alteration of reality via fiber modification

Now suppose we allow ontological non-closure–that is, an entity external to spacetime may alter which possibilities are realized. We can now see that spacekime allows a mathematically precise description of a "reality dial". This "reality dial" acts only on the imaginary-time fiber, not directly on spacetime dynamics. Its actions are of two distinct kinds. First, the dial may restrict the admissible imaginary-time support:

$$\mathcal{T}^*(x,t); \rightarrow; \mathcal{T}^{*'}(x,t) \subset \mathcal{T}_{(x,t)}.$$

This controls how much modal coherence exists, as well as the strength and speed of collapse or decoherence. Support restriction alone reduces coherence but does not select a unique outcome unless only one b remains. Second, the dial may impose boundary conditions on the Schrödinger bridge process in imaginary time:

$$\Phi(\cdot, b = \beta) \in \mathcal{C}_{\text{dial}}.$$

This introduces asymmetry among the remaining possibilities and selects which branch is realized. In spacetime, this appears as non-Hermitian suppression of incompatible components.

2.5 Effective non-Hermitian interface

The spacetime-facing hybrid state can be written in non-Hermitian Hamiltonian form

$$\hat{H}_{\text{eff}} = \hat{H}_1 + i\hat{H}_2, \qquad \hat{H}_2 = \hat{H}_L + \hat{H}_E.$$

Here $\hat{H}_1$ is the spacetime (law-like) Hermitian generator, while $\hat{H}_2$ is the Hermitian operator encoding modal admissibility/selection pressure, decomposed into a law-like SQ contribution $\hat{H}_L$ and an external-interface ("reality dial") contribution $\hat{H}_E$.

Equivalently, the spacetime-facing hybrid state evolves as

$$\partial_a|\Psi\rangle - \frac{i}{\hbar}\hat{H}_1|\Psi\rangle + \frac{1}{\hbar}\hat{H}_2|\Psi\rangle + (\text{noise}).$$

The norm $W(a)$ may change, but the normalized physical state $|\psi_{\text{phys}}(a)\rangle$ remains well defined. Eigenvalues of the effective generator now have the form

$$\lambda_n = E_n - \frac{i}{2}\gamma_n.$$

The eigenvalues of Hermitian generators are conventionally interpreted as energy states. But how should we interpret these complex eigenvalues? The real parts E_n are physical energies (spacetime-real). The imaginary parts γ_n are rates of inadmissibility / selection pressure–how strongly the modal sector suppresses that component given the constraints of the imaginary time Schrödinger bridge and the reality dial.

2.6 Summary

1. Imaginary time forms a finite, quantized fiber enumerating possible physical realizations at each spacetime point.
2. Possibilities are equally weighted but correlated, not independent.
3. Local coherence, maximum entropy, and constraint-driven suppression uniquely select stochastic quantization as the admissible fiber structure.
4. The wavefunction is the coherent synthesis over the SQ-constrained fiber.
5. The Schrödinger equation emerges as the minimal real-time dynamics preserving coherence of possibilities in imaginary time.
6. An external entity may alter reality by restricting fiber support and/or imposing boundary conditions, inducing non-Hermitian effects while preserving normalized physical states.

3 Wavefunction Collapse Example in Single Qubit System

To make the spacekime framework more concrete, we now work out a specific example showing (i) the fiber contributions $\Phi(b)$ under an SQ constraint, and (ii) how a "reality dial" change appears in spacetime as an external-interface contribution $\hat{H}_E$ inside a non-Hermitian Hamiltonian $\hat{H}_1 + i\hat{H}_2$, describing continuous collapse of a single qubit.

3.1 Spacekime Qubit Setup

Let $\mathcal{H} = \text{span}\{|0\rangle, |1\rangle\}$ and consider a pre-measurement physical state

$$|\psi(0)\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

We represent the spacetime-facing (possibly unnormalized) state by $|\Psi(a)\rangle$, with

$$W(a) = \langle\Psi(a)|\Psi(a)\rangle, \qquad |\psi_{\text{phys}}(a)\rangle = \frac{|\Psi(a)\rangle}{\sqrt{W(a)}} \quad (W > 0).$$

Fix a real-time slice a (i.e., $a = \Re(t)$ for $t = a + ib$). The imaginary-time fiber is

$$\mathcal{T} = 0, \Delta b, 2\Delta b, \ldots, \beta, \qquad N := \beta/\Delta b.$$

Each $b_j = j\Delta b$ is equally weighted a priori:

$$\mathbb{P}(b_j) = \frac{1}{N+1}.$$

The possibilities are not independent, because SQ imposes local coherence and maximum entropy correlations along b.

At each b_j let the contribution be a qubit vector

$$\Phi_j := \Phi(b_j) = \alpha_j|0\rangle + \beta_j|1\rangle \in \mathcal{H}.$$

3.2 Imaginary-Time Schrodinger Bridge for Single Qubit

To keep the example explicit and solvable, we choose the maximum-entropy local-coherence prior to be discrete Brownian increments in b:

$$\alpha_{j+1} = \alpha_j + \sqrt{2\nu\Delta b}\,\xi_j^\alpha, \qquad \beta_{j+1} = \beta_j + \sqrt{2\nu\Delta b}\,\xi_j^\beta,$$

with $\xi_j^\alpha, \xi_j^\beta$ i.i.d. standard complex Gaussians (this is the discrete Wiener reference).

Now impose constraint-driven suppression with a Euclidean penalty functional that discourages amplitude in $|1\rangle$:

$$S_E[\{\alpha_j, \beta_j\}] = \frac{\kappa}{2}\sum_{j=0}^{N} |\beta_j|^2\,\Delta b.$$

Here $\kappa \geq 0$ is a constraint strength, and $\kappa = 0$ gives a pure maximum-entropy Brownian fiber (no preferred outcome), while $\kappa > 0$ suppresses $|1\rangle$ across the fiber. SQ says the joint law over the entire fiber $\{\Phi_j\}$ is the least structured (max-entropy) law consistent with local increments and the penalty S_E.

In the most probable path (saddle-point / zero-noise) approximation, this produces a deterministic relaxation of β_j along b:

$$\beta_{j+1} - \beta_j \approx -\kappa\, \beta_j\, \Delta b,$$

so

$$\boxed{\beta(b) \approx \beta(0)\, e^{-\kappa b}.} \tag{B.1}$$

Meanwhile, for this minimal example we keep $\alpha(b) \approx \alpha(0)$ (no penalty on $|0\rangle$).

This is the simplest explicit SQ-shaped fiber: local coherence + maximum entropy prior + constraint-driven suppression yields an exponential decay of the disfavored branch across b.

The spacetime wavefunction at real time a is then the coherent synthesis over the fiber:

$$\boxed{|\Psi(a)\rangle := \sum_{j=0}^{N} \Phi_j(a)\, \Delta b \approx \int_0^{\beta} \Phi(a,b)\, db.} \tag{C.1}$$

In the deterministic approximation above,

$$\Phi(a,b) \approx \alpha(0,a)|0\rangle + \beta(0,a)e^{-\kappa b}|1\rangle.$$

Then the fiber integral is explicit:

$$|\Psi(a)\rangle \approx \left(\int_0^{\beta} \alpha(0,a)\, db\right)|0\rangle + \left(\int_0^{\beta} \beta(0,a)e^{-\kappa b}\, db\right)|1\rangle,$$

so

$$\boxed{|\Psi(a)\rangle \approx \beta\, \alpha(0,a)|0\rangle + \frac{1-e^{-\kappa\beta}}{\kappa}\, \beta(0,a)|1\rangle.} \tag{C.2}$$

If $\kappa = 0$, the $|1\rangle$ term contributes $\beta\, \beta(0,a)$ (no suppression). If $\kappa\beta \gg 1$, the $|1\rangle$ term is suppressed by $\sim 1/\kappa$, while the $|0\rangle$ term is $\sim \beta$. Thus the dial-controlled SQ penalty κ biases which basis component survives the fiber synthesis. Finally, the physical (normalized) state is always

$$|\psi_{\text{phys}}(a)\rangle = \frac{|\Psi(a)\rangle}{\sqrt{\langle\Psi(a)|\Psi(a)\rangle}}.$$

3.3 Reality Dial as Imaginary Time Constraints

Now suppose we let the "reality dial" act at some real time $a_\star$ (i.e., on the $a = \Re(t)$ axis) by changing SQ constraints in the fiber. There are two minimal dial actions: support restriction and boundary conditioning. If the dial reduces imaginary time support β, then fewer b-quanta contribute. This controls collapse speed/strength but does not pick an outcome by itself unless only one b remains. The reality dial can

also impose a terminal boundary condition that makes one branch admissible and the other inadmissible at $b = \beta$. In this model, "select outcome 0" is enforced by

$$\beta(b = \beta) = 0 \quad \text{(or very small)}.$$

A minimal way to implement this in SQ is to add a boundary penalty

$$S_{\text{bdry}} = \frac{\kappa_{\text{bdry}}}{2} |\beta(b = \beta)|^2, \qquad \kappa_{\text{bdry}} \gg 1,$$

which forces $\beta(b = \beta) \to 0$ as $\kappa_{\text{bdry}} \to \infty$. This is a Schrödinger-bridge-style terminal constraint. In short, the dial doesn't just shrink the number of b's; it can also enforce which branch survives at the boundary.

3.4 Reality Dial as Non-Hermitian Operator

Now we can connect the fiber story to the spacetime effective dynamics by writing the effective generator in the form

$$\hat{H}_{\text{eff}} = \hat{H}_1 + i\hat{H}_2, \qquad \hat{H}_2 = \hat{H}_L + \hat{H}_E.$$

In terms of $a = \Re(t)$, the (possibly unnormalized) spacetime-facing state obeys

$$\boxed{\frac{d}{da}|\Psi(a)\rangle - \frac{i}{\hbar}\hat{H}_1|\Psi(a)\rangle + \frac{1}{\hbar}\hat{H}_2(a)|\Psi(a)\rangle,} \tag{E.1}$$

where $\hat{H}_L$ encodes law-like SQ constraints and $\hat{H}_E$ encodes the external interface (the dial). This is equivalent to the earlier Γ-notation via $\hat{H}_2 := -\Gamma/2$.

To match the "collapse to $|0\rangle$" dial action, choose the minimal diagonal external-interface operator

$$\boxed{\hat{H}_E(a) = -\frac{\gamma_{\text{dial}}(a)}{2}\,\Pi_1 = -\frac{\gamma_{\text{dial}}(a)}{2}\,|1\rangle\langle 1|.} \tag{E.2}$$

This yields (for $\hat{H}_1 = 0$ and $\hat{H}_L = 0$) component equations

$$\dot{\alpha}(a) = 0, \qquad \dot{\beta}(a) = -\frac{\gamma_{\text{dial}}(a)}{2\hbar}\beta(a),$$

so

$$\boxed{\beta(a) = \beta(a_\star)\exp\left(-\frac{1}{2\hbar}\int_{a_\star}^{a}\gamma_{\text{dial}}(s)\,ds\right), \qquad \alpha(a) = \alpha(a_\star).} \tag{E.3}$$

After conditioning,

$$|\psi_{\text{phys}}(a)\rangle = \frac{\alpha(a)|0\rangle + \beta(a)|1\rangle}{\sqrt{|\alpha(a)|^2 + |\beta(a)|^2}} \longrightarrow |0\rangle \quad \text{as } \beta(a)/\alpha(a) \to 0.$$

So $\hat{H}_E$ is exactly the spacetime shadow of "reality dial selects outcome 0."

The SQ penalty κ suppresses $\beta(b)$ across imaginary time:

$$\beta(b) \approx \beta(0)e^{-\kappa b}.$$

When you synthesize into spacetime by integrating over $[0, \beta]$, the effective contribution of the $|1\rangle$ branch is reduced by a factor

$$\frac{1 - e^{-\kappa\beta}}{\kappa} \sim \frac{1}{\kappa} \quad (\kappa\beta \gg 1).$$

When the dial changes κ and/or imposes the boundary penalty $\beta(b = \beta) \approx 0$, the fiber synthesis immediately becomes dominated by the $|0\rangle$ branch. In spacetime language, that dominance is equivalently described by turning on $\hat{H}_E \propto -\Pi_1$, which exponentially suppresses $\beta(a)$ along the real-time axis a.

A clean rate matching identification (at the level of timescales) is:

$$\boxed{\text{imaginary-time suppression: } b_{\text{sup}} \sim \kappa^{-1} \qquad \leftrightarrow \qquad \text{real-time collapse: } a_{\text{coll}} \sim \frac{\hbar}{\gamma_{\text{dial}}}.} \tag{F.1}$$

The dial can tune collapse by tuning either side of this correspondence: changing the SQ constraint strength κ (modal side) or the effective γ_{dial} (spacetime side).

3.5 Summary

1. $\Phi(b)$: in the minimal SQ model, each fiber contribution is a qubit vector $\alpha(b)|0\rangle + \beta(b)|1\rangle$ with local coherence in b; constraint-driven suppression yields $\beta(b) \propto e^{-\kappa b}$.
2. Wavefunction synthesis: the spacetime state is the coherent integral/sum over b, which gives an explicit expression (C.2) showing how the disfavored component is suppressed as $\kappa\beta$ grows or as boundary conditioning forces $\beta(b = \beta) \to 0$.
3. Non-Hermitian generator: the same dial choice is equivalently represented in spacetime by turning on $\hat{H}_E = -(\gamma_{\text{dial}}/2)\Pi_1$, which suppresses the $|1\rangle$ amplitude exponentially along a (E.3).
4. Normalization: $|\Psi\rangle$ may change norm, but the physical state $|\psi_{\text{phys}}\rangle$ is always normalized by conditioning.

4 Complex-Time Schrodinger Dynamics and Lindblad Open Quantum System Models

Lindblad equations are often introduced as the standard effective description of noisy quantum systems, where interaction with an environment causes decoherence and dissipation [6–8]. In this paper, however, they are not taken as the starting point. Instead, we begin from a richer complex-time picture in which quantum states evolve on imaginary-time fibers and undergo constraint-driven thinning. The central question of this section is therefore conceptual: when does that deeper fiber-level dynamics look,

from ordinary spacetime, like the familiar Lindblad evolution used in open-system quantum mechanics?

The theorem below provides this bridge. It shows that if thinning events are sufficiently small, numerous, and weakly correlated at coarse-grained timescales, then their averaged effect reduces to a completely positive, trace-preserving Markovian flow of GKSL/Lindblad form. In this sense, Lindblad dynamics appears as an emergent macroscopic limit rather than a fundamental law. This perspective also clarifies scope: when these assumptions fail (for example, due to long memory or strong event correlations), the complex-time model naturally predicts non-Markovian behavior beyond standard Lindblad descriptions.

4.1 Lindblad as Coarse-Grained Limit of Imaginary-Time Thinning

Theorem 1 *Let $\mathcal{H}$ be a system Hilbert space. Suppose the spacekime model assigns to each real-time coordinate $a = \Re(t)$ an imaginary-time fiber $b \in [0, \beta(a)]$ with a fiber field $\Phi(b, a) \in \mathcal{H}$. Define the spacetime-facing state*

$$|\Psi(a)\rangle := \int_0^{\beta(a)} w(b, a)\, |\Phi(b, a)\rangle\, db, \qquad |\psi_{phys}(a)\rangle := \frac{|\Psi(a)\rangle}{\sqrt{\langle\Psi(a)|\Psi(a)\rangle}},$$

for some nonnegative weight function w. Assume:

(A1) For each fixed a, fiber evolution in b is generated by an SQ Markov process whose infinitesimal generator induces a linear map on the projected density matrix after marginalization over b.

(A2) Changes in support $\beta(a)$ occur via many small constraint events at real-time instants a_n, each implementing a completely positive (CP) instrument $\{\mathcal{J}_\mu\}_\mu$ on $\mathcal{B}(\mathcal{H})$, with nonselective map $\sum_\mu \mathcal{J}_\mu$ trace-preserving.

(A3) There exists a coarse-graining scale Δa that is large compared to microscopic SQ-noise correlation times and small compared to macroscopic system evolution times, so event statistics are approximately stationary within each interval.

(A4) In each interval $[a, a + \Delta a]$, event occurrences are conditionally independent given the current projected state and obey Poisson statistics with rates $\gamma_k(a)$ (possibly slowly varying in a).

(A5) Each event produces an $O(\Delta a)$ change in the ensemble state, enabling a first-order expansion in Δa.

(A6) Between events, projected real-time evolution is generated by the Hermitian spacetime operator $\hat{H}_1$.

Then the ensemble-averaged projected density matrix

$$\rho_{phys}(a) := \mathbb{E}\big[|\psi_{phys}(a)\rangle\langle\psi_{phys}(a)|\big]$$

obeys, in the limit $\Delta a \to 0$, a Lindblad master equation

$$\frac{d\rho_{phys}}{da} = -\frac{i}{\hbar}[\hat{H}_1, \rho_{phys}] + \sum_k \left(\hat{L}_k \rho_{phys} \hat{L}_k^\dagger - \frac{1}{2}\{\hat{L}_k^\dagger \hat{L}_k, \rho_{phys}\}\right), \qquad \hat{L}_k := \sqrt{\gamma_k}\, M_k,$$

for operators M_k determined by the first-order Kraus expansion of the constraint-event instrument.

Proof sketch **Step 1: Represent constraint-thinning as a quantum instrument.** By (A2), a single constraint event has outcomes μ with CP maps

$$\mathcal{J}_\mu(\rho) = \sum_\alpha K_{\mu\alpha}\rho K_{\mu\alpha}^\dagger.$$

The nonselective map $\mathcal{J} := \sum_\mu \mathcal{J}_\mu$ is trace-preserving [9]:

$$\sum_{\mu,\alpha} K_{\mu\alpha}^\dagger K_{\mu\alpha} = I.$$

Step 2: Write no-event conditional evolution in complex-time notation. In $[a, a+\Delta a]$, suppose either no event occurs or exactly one event of type k occurs (multi-event probability is $o(\Delta a)$ by (A4)). Let K_k denote event-channel Kraus operators. For Poisson statistics,

$$\Pr(\text{event } k \text{ in } \Delta a) = \gamma_k(a)\Delta a + o(\Delta a), \qquad \Pr(\text{no event}) = 1 - \sum_k \gamma_k(a)\Delta a + o(\Delta a).$$

The no-event branch may be written

$$|\Psi(a+\Delta a)\rangle_{\text{no}} = \left(I - \frac{i}{\hbar}\hat{H}_1\Delta a - \frac{1}{2}\sum_k \gamma_k(a)K_k^\dagger K_k\,\Delta a\right)|\Psi(a)\rangle + o(\Delta a),$$

so the conditional generator is

$$\hat{H}_{\text{eff}}(a) = \hat{H}_1 + i\hat{H}_2^{\text{jump}}(a), \qquad \hat{H}_2^{\text{jump}}(a) := -\frac{\hbar}{2}\sum_k \gamma_k(a)K_k^\dagger K_k.$$

This matches the earlier decomposition $\hat{H}_{\text{eff}} = \hat{H}_1 + i\hat{H}_2$, with the jump-induced dissipative contribution absorbed into $\hat{H}_2 = \hat{H}_L + \hat{H}_E$.

Step 3: Average over events. The ensemble update over Δa is

$$\rho_{\text{phys}}(a+\Delta a) = \Pr(\text{no})\,\rho_{\text{no}}(a+\Delta a) + \sum_k \Pr(k)\,\rho_k(a+\Delta a) + o(\Delta a).$$

For the no-event channel,

$$\rho_{\text{no}}(a+\Delta a) = \frac{A\rho_{\text{phys}}(a)A^\dagger}{\text{Tr}(A\rho_{\text{phys}}A^\dagger)}, \qquad A = I - \frac{i}{\hbar}\hat{H}_1\Delta a - \frac{1}{2}\sum_k \gamma_k(a)K_k^\dagger K_k\,\Delta a,$$

and multiplying by $\Pr(\text{no}) \approx \text{Tr}(A\rho_{\text{phys}}A^\dagger)$ gives

$$\Pr(\text{no})\,\rho_{\text{no}}(a+\Delta a) \approx A\rho_{\text{phys}}A^\dagger.$$

For event k,

$$\rho_k(a+\Delta a) = \frac{K_k\rho_{\text{phys}}(a)K_k^\dagger}{\text{Tr}(K_k\rho_{\text{phys}}K_k^\dagger)}, \qquad \Pr(k) \approx \gamma_k(a)\Delta a\,\text{Tr}(K_k\rho_{\text{phys}}K_k^\dagger),$$

hence

$$\Pr(k)\,\rho_k(a+\Delta a) \approx \gamma_k(a)\Delta a\,K_k\rho_{\text{phys}}K_k^\dagger.$$

Therefore,

$$\rho_{\text{phys}}(a+\Delta a) \approx A\rho_{\text{phys}}A^\dagger + \sum_k \gamma_k(a)\Delta a\,K_k\rho_{\text{phys}}K_k^\dagger.$$

Expanding $A\rho_{\text{phys}}A^\dagger$ to first order gives

$$A\rho_{\text{phys}}A^\dagger = \rho_{\text{phys}} - \frac{i}{\hbar}[\hat{H}_1, \rho_{\text{phys}}]\Delta a - \frac{1}{2}\sum_k \gamma_k(a)\{K_k^\dagger K_k, \rho_{\text{phys}}\}\Delta a + o(\Delta a),$$

so

$$\frac{\rho_{\text{phys}}(a+\Delta a)-\rho_{\text{phys}}(a)}{\Delta a} = -\frac{i}{\hbar}[\hat{H}_1, \rho_{\text{phys}}] + \sum_k \gamma_k(a)\left(K_k \rho_{\text{phys}} K_k^\dagger - \frac{1}{2}\{K_k^\dagger K_k, \rho_{\text{phys}}\}\right) + o(1).$$

Taking $\Delta a \to 0$ yields GKSL form. □

4.2 Interpretation and Scope

If assumptions (A2)–(A6) hold, then non-Hermitian complex-time dynamics is the conditional (no-jump) sector in a, while stochastic thinning plus coarse-graining yields standard open-system Lindblad evolution for $\rho_{\text{phys}}(a)$.

If thinning events are correlated across the fiber or retain memory, assumption (A4) fails, and the reduced dynamics may become time-nonlocal in a, non-GKSL, or even non-CP.

Proposition 2 (Near-Markov limit) *If constraint correlations decay faster than Δa, then coarse-grained reduced dynamics converges to a GKSL semigroup. If they decay more slowly than Δa, the reduced dynamics is generically non-Markovian and non-Lindbladian.*

Conceptually, process memory appears when the effective non-Hermitian generator cannot be written as a purely instantaneous loss term with rates determined only by the present reduced state. In our notation, this means the anti-Hermitian sector $\hat{H}_2(a)$ (or equivalently $\hat{H}_{\text{eff}}(a)$) must retain dependence on unresolved fiber variables and past constraint history. Operationally, memory is expected when these channels have long correlation times, when different-time dissipative generators fail to commute, or when delayed feedback re-injects information from previously suppressed branches. In such cases, the no-jump evolution itself carries history, and future thinning depends on the trajectory, not only on $\rho_{\text{phys}}(a)$.

This perspective is especially relevant for current quantum computers, where internal observers often cannot fully model all environmental pathways in real time. Crosstalk, control-latency feedback, drift, two-level defects, and colored noise can generate correlations that violate Markov assumptions while remaining partially hidden from the device-level model. Lindblad descriptions are still valuable as effective approximations, but they may require ad hoc time-dependent rates or enlarged ancilla spaces to capture long-memory behavior [8]. In the complex-time framework, these hard-to-model channels are absorbed into history-dependent contributions to the non-Hermitian sector, providing a principled route to represent coherent error memory and non-Markovian open-system effects beyond standard GKSL closure.

5 Gravity in Complex Time: A Continuous Collapse Semiclassical Theory of Quantum Gravity

In summary, the complex-time Schrödinger dynamics derived above describe an open quantum system with continuous wavefunction collapse. This suggests a novel

approach to quantum gravity in which the semiclassical Einstein equations still apply to partially collapsed wavefunctions, reducing to the standard equations of relativity in the limit of complete collapse [10]. Two intuitions guide our approach here. First, a partially collapsed wavefunction *should still gravitate*; the idea that gravity only arises from fully collapsed wavefunctions comes from the restrictions of instantaneous wavefunction collapse and unitary dynamics. These assumptions are no longer mathematically required in complex-time Schrödinger dynamics. A second intuition is that once we introduce complex time into the Schrödinger equation, we need to update the semiclassical Einstein equations to account for a five-dimensional stress-energy tensor. Proper gravitational bookkeeping requires accounting for not just completely collapsed wavefunctions but all live possibilities that remain consistent with partially collapsed wavefunctions. Based on these insights, we formulate a collapse-consistent semiclassical theory of quantum gravity in a five-dimensional spacekime manifold consisting of spacetime plus a quantized imaginary-time fiber. The theory is defined by the following postulates.

5.1 Postulate I: Spacekime Structure

There exists an imaginary-time fiber bundle at each spacetime location. As described in Section 4, the imaginary-time fiber is finite and quantized.

5.2 Postulate II: Complex-Time Quantum Dynamics

Quantum dynamics in spacekime are governed by a complex-time extension of the Schrödinger equation as described in previous sections. The wavefunction arises from the integral of a stochastic quantization process over the imaginary-time axis. Continuous wavefunction collapse can be described by an effective non-Hermitian Hamiltonian.

5.3 Postulate III: 5D Stress-Energy Conservation

There exists a five-dimensional stress-energy tensor $\mathbb{T}_{AB}(x, b)$ satisfying

$$\nabla^A \mathbb{T}_{AB} = 0.$$

The effective four-dimensional stress-energy tensor sourcing gravity accounts for the full support over the imaginary-time axis, obtained by integrating over b:

$$T^{\text{eff}}_{\mu\nu}(x) = \sum_b w(x, b) \, \langle T_{\mu\nu}(x; b) \rangle.$$

The weights come from the stochastic quantization process described above. Appropriate boundary conditions on the stochastic quantization process ensure that the net b-flux through the fiber boundaries vanishes globally, guaranteeing that the 4D Bianchi identity is preserved:

$$\nabla^\mu T^{\text{eff}}_{\mu\nu} = 0.$$

This postulate is the direct extension of local energy-momentum conservation from 4D spacetime to 5D spacekime. In general relativity, local conservation is encoded

by $\nabla^\mu T_{\mu\nu} = 0$; here we impose the analogous covariant law on the full spacekime manifold, allowing transport not only across spacetime directions but also along the imaginary-time fiber b. This is also consistent with relativistic quantum continuity principles (for example, the Dirac current conservation law), now interpreted as part of a higher-dimensional conserved flow whose 4D projection gives the physical source seen by spacetime observers.

5.4 Postulate IV: Gravitational Field Equation

The spacetime metric $g_{\mu\nu}(x)$ satisfies the semiclassical Einstein equation with the effective source:

$$G_{\mu\nu}[g](x) = 8\pi G_N T^{\text{eff}}_{\mu\nu}(x).$$

where $T^{\text{eff}}_{\mu\nu}$ is defined in the previous postulate and the cosmological constant is included implicitly in $T^{\text{eff}}_{\mu\nu}$.

Note that the 5D geometry is not assumed to be dynamical; the gravitational dynamics remain 4D. In the limit of complete imaginary-time thinning (complete wavefunction collapse; support concentrated at $b = \Delta b$), the theory reduces to general relativity sourced by a single realized branch.

5.5 Postulate V: Imaginary Time as Nonclassical Mediator of Gravitational Entanglement

The imaginary-time fiber constitutes a nonclassical mediator of gravitational entanglement with a finite Hilbert space dimension determined by $N_{\max}$.

Previous proposals by Bose, Marletto, and Vedral (BMV) argued that if two masses become entangled only via their mutual gravitational interaction, then the mediator must have "quantum-like" features [11, 12]. A formalization of this intuition appears in no-go and classical-channel analyses: under certain assumptions (gravity mediates + gravity is classical + standard subsystem locality), entanglement generation is impossible [12, 13]. Consequently, any model claiming gravitational entanglement must violate at least one assumption behind those theorems. The violation in our case is that the imaginary-time fiber constitutes a nonclassical mediator. The fiber may encode non-separable constraints on joint configurations of spatially separated systems. Thus, our theory predicts the possibility of gravitational entanglement without contradicting established no-go theorems.

5.6 Postulate VI: Unified Dark Sector

Consistent with the detailed construction in Section 7 (*Dark Matter and Dark Energy*), sector assignments are determined by the coarse-grained support variable Θ: intermediate support $\Theta_{\min} < \Theta < \Theta_{\max}$ contributes a dust-like component (dark matter), fully thinned local states $\Theta = \Theta_{\min}$ correspond to baryonic matter, and an irreducible homogeneous residual support contributes a vacuum-like component (dark energy).

The effective equation-of-state parameter evolves as

$$w_{\text{eff}}(a) = -\frac{\rho_{\text{vac}}}{\rho_{\text{DM}} + \rho_{\text{vac}}}.$$

During structure formation, the clustering component remains pressureless to leading order, ensuring compatibility with CMB and large-scale structure constraints [14, 15].

5.7 Postulate VII: Entropy Bound and Cosmological Constant

The finite support capacity of the fiber determines the cosmological constant.

Matching fiber entropy to de Sitter horizon entropy yields [16]

$$\Lambda = \frac{3\pi}{\ell_P^2 N_{\max}}.$$

In the Section 7 language, finite support capacity $N_{\max}$ fixes the irreducible modal floor $\Theta_{\min}$. That same non-depletable floor is what appears macroscopically as the homogeneous vacuum-like component.

Thus the cosmological constant and the matter–vacuum transition epoch are controlled by the single structural parameter $N_{\max}$.

5.8 Gravitational Entanglement Effects

To demonstrate the possibility of gravitationally mediated entanglement, we work out the details of a BMV-style branch-qubit toy system. Two masses in superposition are modeled in with initial state $|+\rangle_A|+\rangle_B$. A fiber-mediated controlled phase $U_{\text{eff}} = \text{diag}(1, 1, 1, e^{i\phi})$ gives coherent-limit concurrence $C_{\text{pure}} = |\sin(\phi/2)|$. Continuous thinning is then represented as branch-basis dephasing $r(t) = e^{-\Gamma t}$, which yields the positivity condition $r(t)|\sin(\phi/2)| > (1 - r(t))/2$. For the maximally entangling setting $\phi = \pi$ reached at $t_{\text{gate}} = \pi/\Omega_{\text{sk}}$, this implies $\Gamma/\Omega_{\text{sk}} < \ln 3/\pi$. Therefore,

$$\Gamma < \frac{\ln 3}{\pi}\,\Omega_{\text{sk}} \approx 0.35\,\frac{\chi\,\Delta U}{\hbar}, \qquad \Omega_{\text{sk}} := \frac{\chi\,\Delta U}{\hbar}.$$

Here ΔU is the branch-dependent gravitational energy difference, $\chi \in (0, 1]$ quantifies coherent (b)-participation in the fiber-mediated phase channel, and Γ is the effective thinning/dephasing rate induced by the non-Hermitian generator. Intuitively, this derivation shows that gravitationally-mediated entanglement is possible under certain conditions on the imaginary-time thinning rate.

Finally, note that the b-sector is nonlocal but not operationally controllable from within spacetime; thus the theory preserves no-signaling at the level of observable spacetime dynamics.

5.9 Explaining the Page-Geilker Experiment

This new collapse-consistent semiclassical theory of gravity avoids a long-identified pitfall with semiclassical theories of quantum gravity. Page and Geilker's 1981 Cavendish-style "Schrodinger's-cat" experiment couples a quantum-triggered macroscopic mass rearrangement (two near-equiprobable configurations, AB versus BA) to a torsion balance [17]. In naive semiclassical gravity [10],

$$G_{\mu\nu} = 8\pi G_N \langle T_{\mu\nu} \rangle,$$

the balance should respond to the Born-weighted average mass distribution, implying a weak and largely uncorrelated torque. Instead, the observed deflection is strongly correlated with the realized configuration in each run, ruling out that average-source prediction.

In spacekime gravity, the metric is sourced by the imaginary-time marginal effective tensor,

$$T^{\text{eff}}_{\mu\nu}(x) = \sum_b w(x,b) \langle T_{\mu\nu}(x;b) \rangle.$$

Macroscopic amplification drives rapid imaginary-time thinning, concentrating the weight function w onto the realized branch. At the torsion-balance level, this gives a continuous interpolation from the semiclassical average torque to the branch-conditioned torque:

$$\tau(t) \approx (1-\eta(t))\tau_{\text{avg}} + \eta(t)\tau_{\text{branch}}, \qquad \eta(t) = 1 - \exp\left(-\frac{\|H_E\|}{\hbar} t\right),$$

where τ_{avg} is the torque from the averaged mass distribution (approximately canceling for near-equal Born weights), τ_{branch} is the torque for the realized configuration (AB or BA), and the non-Hermitian thinning generator is decomposed as $H_2 = H_L + H_E$, with H_L the law-like background channel and H_E the external constraint/measurement channel that drives branch selection.

In the Page–Geilker regime, the characteristic H_E-driven thinning timescale $\hbar/\|H_E\|$ is much shorter than the torsion-balance response and averaging time, so $\eta \to 1$ and $\tau(t) \to \tau_{\text{branch}}$. This yields the observed near-perfect branch correlation while preserving 4D conservation as the b-projection of 5D conservation closure.

5.10 Proposed Experiments for Testing the Theory

Our theory is testable in the near term using recently proposed experiments. A first direct experimental route is provided by postselected weak-value protocols that convert gravity-induced branch phases into anomalously large, and potentially sign-reversed, effective impulses for selected runs [18]. In the spacekime picture, these experiments probe the competition between coherent branch-phase accumulation $\Omega_{\text{sk}} = \chi\,\Delta U/\hbar$ and measurement-driven thinning through the external channel H_E. The concrete prediction is a three-way trade-off between weak-value amplification, postselection success probability, and residual branch coherence: stronger effective branch selection (larger $\|H_E\|$, larger η) should suppress anomalous weak-value signatures through faster thinning. Measuring these observables simultaneously gives direct bounds on $(\chi, \Gamma, \|H_E\|)$ and tests whether gravity follows continuous branch-conditioned collapse rather than a purely unitary phase model or a Born-averaged classical source.

A complementary route is the many-body Schrödinger–Newton optomechanics program, which identifies test observables in mean motion, covariance dynamics, and interferometric visibility, with realistic near-term sensitivity estimates [19]. In our framework, these channels are fit by a common effective source $T^{\text{eff}}_{\mu\nu}$ together with

thinning-induced non-Hermitian corrections. This yields a falsifiable structure: coherent semiclassical shifts track mass and geometry, while thinning terms track constraint history and environmental activation of H_E. Joint inference across these observables can separate ordinary decoherence from genuine spacekime thinning dynamics, and can directly test our no-signaling claim by verifying that any recovered nonlinear corrections remain operationally local.

6 Dark Matter and Dark Energy

The notion that gravity requires proper bookkeeping over the imaginary-time axis suggests an intriguing idea: a "possibility conservation" law analogous to charge or mass conservation. Another intuition is that we can think of quantized imaginary time support as a reservoir of fluid or charge, and thus continuous wavefunction collapse caused by imaginary time thinning is analogous to a fluid flow. Either intuition motivates a 5D conservation law that links baryonic matter, dark matter, and dark energy to complex-time Schrodinger dynamics.

Let $\Theta(x,t)$ denote the depletable imaginary-time support (possibility thickness) at spacetime point (x,t). We define it as the coarse-grained fiber support

$$\Theta(x,t) := \sum_{b\in\mathcal{T}_{(x,t)}} w(x,t;b)\,\Delta b,$$

with continuum analogue $\Theta(x,t) = \int_0^{\beta(x,t)} w(x,t;b)\,db$. Our key insight is the following identification:

- Dark matter = intermediate imaginary-time support $\Theta_{\min} < \Theta < \Theta_{\max}$
- Baryonic matter = fully thinned state $\Theta = \Theta_{\min}$
- Dark energy = irreducible global residual modal thickness (geometric invariant)

Thus, the dark sector and ordinary matter are not separate substances but different regimes of thinning of the same modal substrate.

6.1 Spacekime Stress-Energy Conservation

As postulated in the previous section:

1. There exists a 5D stress-energy tensor $\mathbb{T}_{AB}(x,b)$ with

$$\nabla^A \mathbb{T}_{AB} = 0.$$

2. The effective 4D source is the fiber marginal:

$$T^{\text{eff}}_{\mu\nu}(x) = \sum_b w(x,b)\,\langle T_{\mu\nu}(x;b)\rangle.$$

3. Vanishing net b-flux ensures

$$\nabla^\mu T^{\text{eff}}_{\mu\nu} = 0,$$

preserving the 4D Bianchi identity.

A partially collapsed wavefunction can consistently gravitate before full collapse. This establishes that incomplete thinning ($\Theta > \Theta_{\min}$) already produces gravitational attraction.

6.2 Conservation of Imaginary-Time Possibility

Applying the same projection logic to the scalar support field $w(x, t; b)$, we introduce a 5D support-flow vector J^A_Θ obeying a continuity law:

$$\nabla_A J^A_\Theta = 0.$$

Intuitively, this describes transport of modal weight ("possibility") in (x, b)-space, analogous to current transport in electromagnetics. The b-component of J^A_Θ quantifies redistribution of admissible support along the fiber, and its boundary term encodes thinning.

Integrating this continuity law over a comoving 3-volume and over the fiber gives

$$\frac{d}{dt}\big(a_{\rm sc}^3\Theta\big) = -\Phi_b,$$

where Φ_b is the net support flux through the fiber boundaries. Under vanishing net fiber-boundary flux ($\Phi_b = 0$), this reduces to

$$\frac{d}{dt}\big(a_{\rm sc}^3\Theta\big) = 0.$$

Thus,

$$H(t) = \frac{1}{a_{\rm sc}}\frac{da_{\rm sc}}{dt} = -\frac{1}{3}\frac{d\ln\Theta}{dt}.$$

Thus, in this model, cosmic expansion is the macroscopic kinematic consequence of thinning imaginary-time support. Combining this with results from the previous section, gravitational attraction and cosmic expansion both arise from conservation laws in spacetime plus imaginary time. There is a structural analogy to ordinary conserved: J^A_Θ is a bookkeeping current on the enlarged spacekime manifold, while only projected quantities such as Θ, $T^{\rm eff}_{\mu\nu}$, and H are directly observable in 4D.

These two conservation statements (conservation of stress-energy and imaginary-time possibility) can be viewed as complementary projections of a single 5D continuity structure on spacekime. The tensor law $\nabla^A \mathbb{T}_{AB} = 0$ governs local energy-momentum transport in the full spacetime-fiber manifold, while the scalar law $\frac{d}{dt}(a_{\rm sc}^3\Theta) = 0$ tracks the comoving budget of surviving modal support after projection onto cosmological variables. Under vanishing net fiber-boundary flux, both descend consistently to 4D: one fixes the effective gravitational source, and the other fixes the thinning-expansion bookkeeping.

6.3 Real-Time Dynamics of Monotonic Imaginary-Time Thinning

Because imaginary time thinning is irreversible (possibilities are only eliminated, never created):

- Θ must decrease monotonically along the realized history.
- No stochastic thickening is allowed.

An Ornstein-Uhlenbeck diffusion process would be a natural choice here, but this would allow violations of monotonic imaginary-time thinning. Thus, we use a one-sided process instead:

$$\Theta(t) = \Theta(0)e^{-\kappa t}(1-\eta)^{N_t}.$$

where

- κ is the continuous thinning rate,
- r is the Poisson jump rate,
- η is the fractional thinning per jump.

This preserves pathwise monotonicity while allowing heterogeneous thinning histories.

This real-time process is the effective coarse-grained image of the imaginary-time stochastic quantization (SQ) dynamics introduced in a previous section. There, the SQ Langevin equation acts along the fiber coordinate b and sets how admissible support is redistributed and suppressed across possibility branches; here, $\Theta(t)$ tracks the projected total surviving support in the cosmological time t. Starting from the previously stated assumptions (local coherence in b, maximum-entropy reference law, and constraint-driven suppression), plus the irreversibility assumption that eliminated support is not regenerated, the induced real-time evolution must be one-sided. Under this projection, continuous suppression contributes the exponential factor $\exp(-\kappa t)$, while intermittent constraint events contribute multiplicative jumps $(1-\eta)^{N_t}$, yielding the monotone thinning law above.

6.4 Deriving the Dark-to-Baryon Ratio

Using the minimal identification:

$$\rho_{\mathrm{DM}} \propto \Theta(t), \qquad \rho_b \propto \Theta(0) - \Theta(t),$$

the observed ratio

$$\frac{\rho_{\mathrm{DM}}}{\rho_b} \approx 5$$

implies

$$\Theta(t_0) = \frac{5}{6}\Theta(0).$$

This yields the constraint:

$$\kappa + r\eta \approx \frac{\ln(6/5)}{t_0} \approx 4 \times 10^{-19}\,\mathrm{s}^{-1} \approx 0.2H_0.$$

The effective thinning-to-floor rate must be of order a fraction of the Hubble rate over cosmic time. This is a nontrivial constraint on the anti-Hermitian Hamiltonian.

6.5 Structure Formation Mechanism

So how do void → halo → galaxy transitions occur?

- Thinning must depend on dissipative interaction channels, not simply on density.
- Gravitational binding alone does not enforce full thinning.
- Full thinning (baryon formation) requires irreversible constraint pathways (cooling, EM interactions).

Thus:

- Voids: slow thinning → high Θ → DM-like.
- Halos: moderate thinning → stable intermediate Θ.
- Galaxies: rapid thinning via dissipation → Θ approaches its minimum quantum.

This matches:

- Collisionless behavior of dark matter in mergers,
- Dense dark matter in galactic centers,
- Dissipative baryon settling.

6.6 Relative Strength of Attraction and Expansion

Define the balance parameter:

$$\Xi(R,t) = \frac{GM_{\text{eff}}(R)}{(\frac{dH}{dt} + H^2)R^3}.$$

- Attraction arises from spatial variation of Θ.
- Expansion arises from global thinning rate.

This connects halo binding, growth rate, and expansion through the same modal variable.

6.7 Dark Energy as Geometric Invariant

The irreducible modal floor $\Theta_{\min}$:

- Cannot be eliminated by thinning,
- Is homogeneous,
- Produces Λ-like behavior.

Dark energy is thus the curvature response to irreducible residual modal thickness, not vacuum energy but geometric inevitability.

This is the same mechanism summarized in Postulate VII: the finite fiber capacity $N_{\max}$ sets the irreducible floor $\Theta_{\min}$, and matching fiber entropy to de Sitter entropy gives $\Lambda = 3\pi/(\ell_P^2 N_{\max})$. Thus the $\Theta_{\min}$ and $N_{\max}$ descriptions are two equivalent parameterizations of one dark-energy source.

6.8 Relation to Particle Dark Matter

Many physicists are currently searching for dark matter particles, including weakly interacting massive particles and axions [20, 21]. Our model is consistent with but does not require such new dark matter particles. There are three possibilities:

1. Purely modal dark matter (no particles).
2. Intermediate Θ manifests as coherent ultra-light field (axion-like).
3. Particles emerge as excitations of partially thinned modal structure.

Ultra-light bosonic dark matter could be the effective field description of intermediate-support modal structure. Thus, detection of axions would not refute our model. However, our model does not require the existence of dark matter particles; the question simply comes down to whether partially thinned imaginary-time support produces particle-like behavior.

The anti-Hermitian generator must produce Hubble-suppressed effective floor-hitting rates. The generator must also likely decompose into a global thinning channel (expansion), and crystallization channel (baryonization). Be selective and constraint-activated (not generic decoherence). Thus the cosmological ratio constrains operator structure.

In summary, attraction, expansion, dark matter, baryons, and dark energy all arise from one modal thickness variable Θ, one non-Hermitian thinning mechanism, and one 5D projection conservation law. This model does not require extra fluids, ad hoc sectors, or arbitrary thresholds. Everything is different regimes of thinning of the same modal substrate. This framework now explains why expansion occurs, explains why dark matter persists, explains baryon minority fraction, explains collisionless behavior, explains late-time acceleration, allows (but does not require) particle dark matter, and produces operator-level constraints.

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